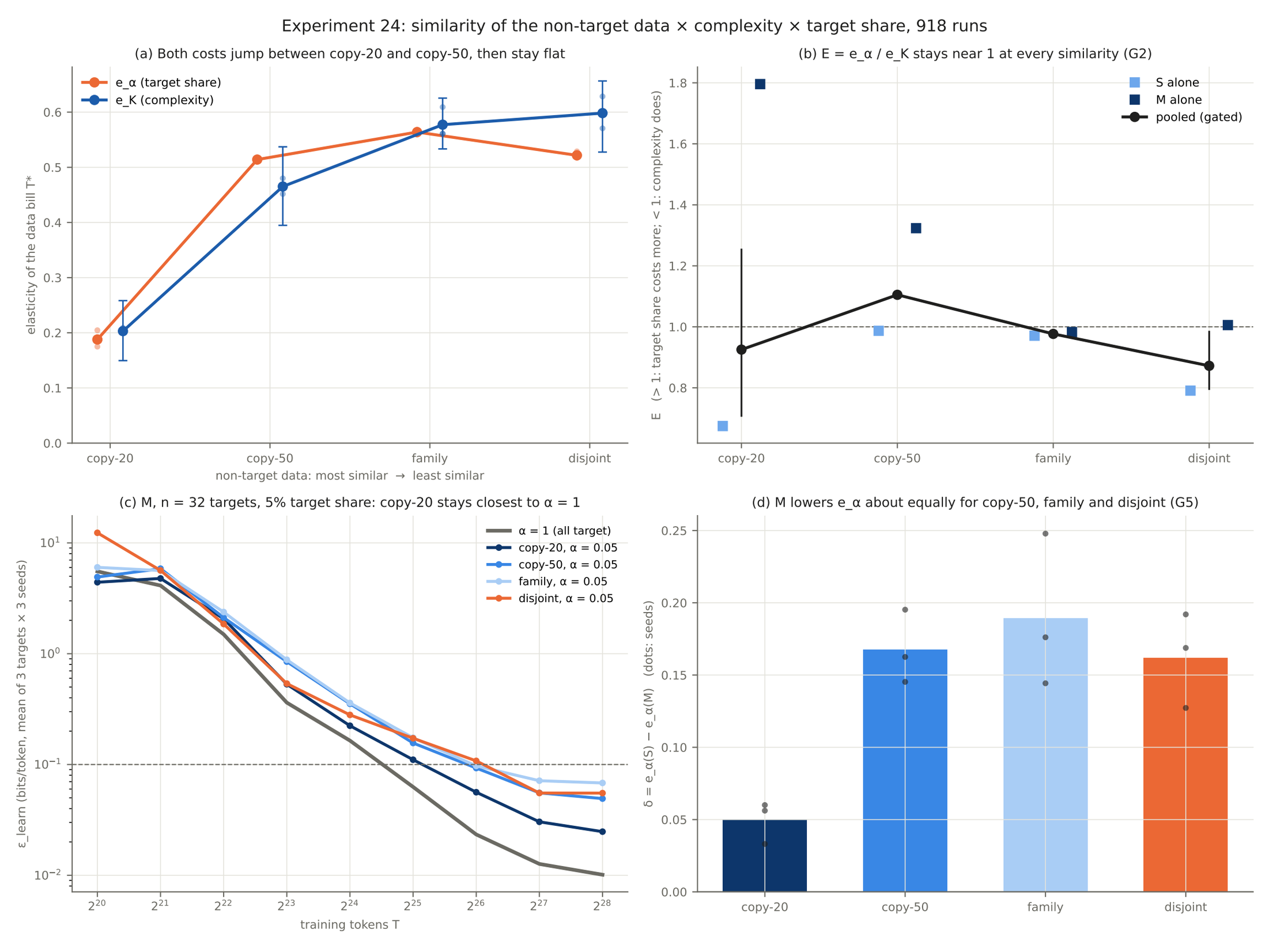
Experiment 24: similarity of the non-target data × complexity × target share, 918 runs
(a) Both costs jump between copy-20 and copy-50, then stay flat
e_α (target share)
e_K (complexity)
elasticity of the data bill T*
0.0
0.1
0.2
0.3
0.4
0.5
0.6
copy-20
copy-50
family
disjoint
non-target data: most similar → least similar
(b) E = e_α / e_K stays near 1 at every similarity (G2)
S alone
M alone
pooled (gated)
E (> 1: target share costs more; < 1: complexity does)
0.8
1.0
1.2
1.4
1.6
1.8
copy-20
copy-50
family
disjoint
(c) M, n = 32 targets, 5% target share: copy-20 stays closest to α = 1
α = 1 (all target)
copy-20, α = 0.05
copy-50, α = 0.05
family, α = 0.05
disjoint, α = 0.05
ε_learn (bits/token, mean of 3 targets × 3 seeds)
10^{1}
10^{0}
10^{-1}
10^{-2}
2^{20}
2^{21}
2^{22}
2^{23}
2^{24}
2^{25}
2^{26}
2^{27}
2^{28}
training tokens T
(d) M lowers e_α about equally for copy-50, family and disjoint (G5)
δ = e_α(S) − e_α(M) (dots: seeds)
0.00
0.05
0.10
0.15
0.20
0.25
copy-20
copy-50
family
disjoint